

FILE TRANSFER PROTOCOL

(FTP)

What is FTP?

- ▶ FTP (File Transfer Protocol) is an Internet protocol used to transfer files between two computers over a TCP/IP network.
- ▶ It allows you to access folders on a remote computer and upload/download files.
- ▶ Local computer → client
Remote computer → server

Why FTP is used?

- ▶ To transfer files between computers
- ▶ To upload website files to a server
- ▶ To download software, images, documents, etc.
- ▶ To share files in businesses
- ▶ To backup files

NOTE

- ▶ FTP hides OS differences, so files can be shared between different systems.

FTP Client and Server Model

- ▶ **FTP Client:** A program that runs on the user's computer, enabling communication with an FTP server. It provides commands to connect, browse and transfer files.
- ▶ **FTP Server:** A system that hosts the files and directories, waiting for requests from clients.

Commands

- ▶ `get filename`(retrieve the file from the directories get server)
- ▶ `mget filename`(retrieve multiple files from the server)
- ▶ `ls`(lists files available in the current directory of the server)

How FTP Works (Simple Steps)

- ▶ Client connects to FTP server on Port 21
- ▶ User logs in (username/password or anonymous)
- ▶ Client can list directories, change folders, request files
- ▶ When a file is transferred, FTP uses a separate data channel (Port 20)
- ▶ File is uploaded/downloaded
- ▶ Control connection stays open until user quits

FTP Channels

1. Command Channel (Port 21)

Used for:

- Login
- Navigation
- Commands

2. Data Channel (Port 20)

Used for:

- Actual file transfer

Types of FTP Connections

1. Active FTP connection

- ▶ In an Active FTP connection, the client establishes the command channel and the server establishes the data channel.
- ▶ When the client requests the data over the connection the server initiates the transfer of the data to the client.
- ▶ It is not the default connection because it may cause problems if there is a firewall in between the client and the server.

2. Passive FTP connection

- ▶ In a Passive FTP connection, the client establishes both the data channel as well as the command channel.
- ▶ When the client requests the data over the connection, the server sends a random port number to the client, as soon as the client receives this port number it establishes the data channel.
- ▶ It is the default connection, as it works better even if the client is protected by the firewall.

Anonymous FTP

Some servers provide anonymous FTP, where files are available for public access without authentication.

- ▶ The username is set to anonymous.
- ▶ The password is typically the user's email address.
- ▶ Access is limited: users can download files but not browse directories or make modifications.

FTP Commands (Basic)

Command	Meaning
get file	Download file
mget file	Download multiple files
put file	Upload a file
ls or dir	List files
cd	Change directory
close	Close connection
quit	Exit FTP session

How FTP Works (Simple Steps)

- ▶ Client connects to FTP server on Port 21
- ▶ User logs in (username/password or anonymous)
- ▶ Client can list directories, change folders, request files
- ▶ When a file is transferred, FTP uses a separate data channel (Port 20)
- ▶ File is uploaded/downloaded
- ▶ Control connection stays open until user quits

Transmission Modes

FTP supports 3 transmission modes:

- ▶ Stream Mode – default, sends data as continuous bytes
- ▶ Block Mode – sends data in blocks
- ▶ Compressed Mode – compresses data to save time & bandwidth

Applications of FTP

- ▶ Business file sharing
- ▶ Backup and recovery
- ▶ Sharing confidential documents
- ▶ Uploading website files
- ▶ Collaborative work

Why FTP?

FTP is reliable because it supports:

- ▶ ASCII and Binary file transfers
- ▶ Compatibility between different operating systems
- ▶ Structured and unstructured file formats